AptitudeX Formula Handbook

A printable quick-revision sheet for placement aptitude

How to use this handbook

Revise one section before practice, then use the AptitudeX question bank for topic-wise drills. Always write the base quantity beside a percentage and normalize units before substituting a speed or work formula.

Formula index

Number System

? HCF ? LCM = product of two numbers

? Sum 1..n = n(n+1)/2

? Divisor ? quotient + remainder = dividend

Percentages

? Percentage = part/whole ? 100

? Successive change = a + b + ab/100

? If A is x% more than B, B is x/(100+x)% less than A

Profit & Loss

? Profit% = (SP?CP)/CP ? 100

? SP = CP(1 + profit%/100)

? Successive discount = d1+d2?d1d2/100

Interest

? SI = PRT/100

? CI = P(1+R/100)^T ? P

? For 2 years, CI?SI = P(R/100)^2

Ratio, Partnership & Mixture

? a:b=c:d ? ad=bc

? Profit share ? capital ? time

? Alligation: cheaper:dearer=(dearer?mean):(mean?cheaper)

Time & Work

? Work rate = 1/days

? Combined rate = sum of individual rates

? Men ? days ? hours = constant

Speed, Trains & Boats

? Speed = distance/time

? km/h to m/s = ?5/18

? Boat=(downstream+upstream)/2

Ages

? Present age ? n = future/past age

? Set ratio ages as kx and apply condition

Permutation, Combination & Probability

? nPr=n!/(n?r)!

? nCr=n!/[r!(n?r)!]

? P(E)=favourable/total

Algebra & Quadratics

? Roots=[?b??(b??4ac)]/2a

? Sum roots=?b/a; product roots=c/a

? (a+b)^2=a^2+2ab+b^2

Geometry & Mensuration

? Triangle area=1/2bh

? Circle area=?r?; circumference=2?r

? Cylinder volume=?r?h

Series, Logs & DI

? log(ab)=log a+log b

? Series: inspect differences, ratios and alternation

? Percentage change=(new?old)/old?100

Clocks & Calendars

? Angle=|30H?11M/2|

? Odd days = total days mod 7

? Leap year: divisible by 4 except non-400 centuries

Mental-math checklist

? Memorise fraction?percentage pairs: 1/2=50%, 1/3=33.33%, 1/4=25%, 1/5=20%, 1/8=12.5%.

? For options, estimate first and eliminate values outside the expected range.

? For work and pipes, add rates; for equal-distance speed, use the harmonic mean.

? For DI, read the units and base year before touching a calculator.